

Advanced Certification in

Applied Data Science, Machine Learning and Edge AI



Electronics & ICT Academy
Indian Institute of Technology Guwahati
An Initiative of Ministry of Electronics & Information Technology (MeitY)



इलेक्ट्रॉनिक्स एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
**ELECTRONICS AND
INFORMATION TECHNOLOGY**

Course Curriculum

Duration: 6 Months (140Hrs)

DS | ML | DL | Edge AI | Generative AI

In Collaboration With

**The IoT
Academy**

Connecting The Unconnected

ABOUT THE IIT GUWAHATI



Indian Institute of Technology Guwahati, the sixth member of the IIT fraternity, was established in 1994. The academic programme of IIT Guwahati commenced in 1995. At present the Institute has eleven departments and three inter-disciplinary academic centres covering all the major engineering, science and humanities disciplines, offering BTech, BDes, MA, MDes, MTech, MSc and PhD programmes. Within a short period of time, IIT Guwahati has been able to build up world class infrastructure and a reputation for itself.

ABOUT

E&ICT ACADEMY, IIT GUWAHATI



Electronics & ICT Academy

Indian Institute of Technology Guwahati

An Initiative of Ministry of Electronics & Information Technology (MeitY)

Electronics and ICT Academy aims to provide specialized training to the faculties of Engineering, Arts, Commerce, Science colleges and Polytechnics institutes by developing short term training programmes on fundamental and advanced topics in IT, Electronics & Communication, Product Design, Manufacturing. In addition, the Academy conducts specialized customized training programmes and research promotion workshops for corporate sector & educational institutions.



Data Analytics, Data Science & Visualization with Python

Module-1: Python Basics

- ✓ What is Python
- ✓ Application of Python
- ✓ Why use Python for AI-ML Installation
Anaconda/Other Idle
- ✓ Python Tokens
- ✓ Data Types in Python
- ✓ Conditional Statement
- ✓ Loops in Python
- ✓ Functions in Python
- ✓ Advance Functions
- ✓ File Handling
- ✓ **Mini Project**

Module-2: Intro to Git & Github and Kaggle

- ✓ Git and GitHub
- ✓ kaggle

Module-3: Python NumPy & Pandas

- ✓ Introduction to NumPy
- ✓ Exploring a NumPy Array
- ✓ Indexing & Slicing a NumPy Array
- ✓ Manipulating a NumPy Array
- ✓ Performing Mathematical & Statistical
Functions using NumPy
- ✓ Performing Linear Algebra Operations
using NumPy
- ✓ Introduction to Pandas
- ✓ Exploring Pandas Series
- ✓ Introduction to Pandas DataFrame
- ✓ Importing & Exporting Data
- ✓ Implementing basic DataFrame
functionalities
- ✓ Exploring Descriptive Statistics with
Pandas
- ✓ **Mini Project**

Modules-4: R Language Essentials

- ✓ Fundamentals of R
- ✓ Vectors & control statements
- ✓ Functions in R
- ✓ Matrices & strings
- ✓ Lists & arrays in R
- ✓ Data visualization in R

Modules-5: Exploratory Data Analysis (EDA)

- ✓ Data Types
- ✓ Dispersion & Skewness
- ✓ Data imputation
- ✓ Data Pre-processing
- ✓ Data Cleaning
- ✓ Data Manipulation
- ✓ Advanced Manipulation
- ✓ **Mini Project**

Module-6: SQL For Data Analytics

- ✓ SQL Basic
- ✓ SQL Joins
- ✓ SQL Aggregations
- ✓ Subqueries and Temp Tables
- ✓ SQL Data Cleaning
- ✓ Window Functions
- ✓ **Mini Project**

Module-7: Data Analysis with Excel

- ✓ Key Formulas and Functions, Ranges and Tables
- ✓ Data Cleaning – Text Functions, Dates and Times
- ✓ Conditional Formatting, Sorting and Filtering
- ✓ Dashboard Creation
- ✓ Analysis with Pivot Tables
- ✓ Data Analysis in Excel – Trends and Patterns
- ✓ Data Visualization in Excel – Charts and Plots
- ✓ Functions in Python
- ✓ Advance Functions
- ✓ File Handling
- ✓ **Mini Project**

Module-8: Data Visualization with Python

- ✓ Why Data Visualization?
- ✓ Introduction to Data Visualization
- ✓ Libraries & Tools for Data Visualization in Python
- ✓ Static Data Visualization Using Seaborn
- ✓ Interactive Data Visualization Using Plotly Express
- ✓ Interactive Animations & Facet Plots

Module-9: Tableau for Business Intelligence

- ✓ Understanding Data
- ✓ Creating Your First visualization
- ✓ Tableau Calculations
- ✓ Formatting Visualizations
- ✓ Manipulating Data in Tableau
- ✓ Creating Dashboards AND Stories
- ✓ Distributing & Publishing Your Visualization
- ✓ **Mini Project**

Module-10: Visual Storytelling using Power BI

- ✓ Introduction To Power BI
- ✓ Creating Power BI Reports, Auto Filters
- ✓ Report Visualization And Properties
- ✓ Chart And Map Report Properties
- ✓ Hierarchies And Drilldown Reports
- ✓ Power BI Deployment & Cloud
- ✓ Improving Power BI Reports And More...
- ✓ **Mini Project**

Capstone Project- 1 in Data Analytics

Applied Maths, Statistics and Probability for ML & AI

Module-1: Mathematics for Machine Learning & AI

- ✓ Linear Algebra
- ✓ Multi-variable Calculus
- ✓ Introduction to Calculus

Module-2: Statistics for Machine Learning & AI

- ✓ Applications of Statistics
- ✓ Sampling techniques
- ✓ Introduction to Statistics
- ✓ Descriptive Statistics
- ✓ Categories of Data
- ✓ Measure Used in Descriptive Statistics
- ✓ Basic Terminologies in Statistics
- ✓ Z-Scores

Module-3: Probability

- ✓ What is Probability?
- ✓ Random Variables
- ✓ Rules of probability
- ✓ Probability Distribution Functions
- ✓ Types of Probability
- ✓ **Mini Projects**

Module-4: Inferential Statistics

- ✓ Introduction to Inferential Statistics
- ✓ One Sample Z test
- ✓ Hypothesis Testing
- ✓ One Sample T test
- ✓ Normal Distribution
- ✓ Independent Sample T test
- ✓ P-value
- ✓ Chi-square test
- ✓ One-tailed and Two-tailed tests
- ✓ ANOVA

Capstone Project- 2

Machine Learning Specialization

Module-1: Introduction to Machine Learning

- ✓ What is Machine Learning?
- ✓ Applications of Machine Learning
- ✓ Machine Learning in your daily life
- ✓ Machine Learning in Retail
- ✓ Steps Involved in Machine Learning

Module-2: Regression

- ✓ Introduction to Regression
- ✓ Linear Regression
- ✓ Evaluation Metrics in Regression Models
- ✓ Logistic Regression
- ✓ Mini Project

Module-3: Supervised Classification

- ✓ Why Use Classification?
- ✓ Application of Classification Algorithms
- ✓ Introduction to Classification
- ✓ Types of Classification Algorithms
- ✓ Classification: Decision Tree
- ✓ Classification: Random Forest
- ✓ ML in Banking & Finance - Benefits
- ✓ Classification: SVM
- ✓ Classification: KNN
- ✓ Classification: Naïve Bayes
- ✓ Evaluating Classification Models
- ✓ Model Optimization Techniques
- ✓ Model Boosting Techniques
- ✓ Introduction to PyCaret
- ✓ Dealing with Unbalanced Datasets
- ✓ 4 Mini projects

Module-4: Unsupervised Learning

- ✓ What is Unsupervised Learning?
- ✓ Application of Unsupervised Learning
- ✓ Introduction to Clustering
- ✓ Types of Clustering
- ✓ Partitioning Methods: K-means, DBSCAN, Spectral
- ✓ Hierarchical Methods: Hierarchical
- ✓ 2 Mini projects

Module 5 :- Dimension Reduction

- ✓ PCA
- ✓ Factor Analysis
- ✓ LDA
- ✓ Mini project

Module-6: Association Rules Mining

- ✓ What are Association Rules?
- ✓ Association Rule Parameters
- ✓ A-priori Algorithm
- ✓ Market Basket Analysis

Module-7: Recommendation System

- ✓ What is a Recommendation System?
- ✓ Need for a Recommendation System
- ✓ Recommendation System Use Cases
- ✓ Applications of Recommendation System
- ✓ Types of Recommendation Systems
- ✓ Collaborative Filtering
- ✓ Content Based Filtering
- ✓ Matrix Factorization
- ✓ Pros and Cons of Collaborative Filtering
- ✓ Content Based filtering
- ✓ Hybrid Recommender System
- ✓ **Mini project**

Module-8: Time-series Forecasting

- ✓ Introduction to forecasting data
- ✓ Properties of Time Series data
- ✓ Features of Time Series data
- ✓ Markov Processes - Overview and Terminologies
- ✓ Naive, Average and Moving Average Forecasting
- ✓ Exponential Smoothing
- ✓ ARIMA Approach
- ✓ **Mini project**

Module-9: Machine Learning Model Deployment

- ✓ Overview of Machine Learning Models
- ✓ Machine Learning System Architecture
- ✓ Research Environment
- ✓ Packaging, Serving and Deploying the model
- ✓ Differential Testing And More...

Deep Learning Basics with TensorFlow & text processing

Module-1: Introduction to Deep Learning

- ✓ What is Deep Learning
- ✓ Curse of Dimensionality
- ✓ Machine Learning vs. Deep Learning
- ✓ Use Cases of Deep Learning
- ✓ Human Brain vs. Neural Network
- ✓ What is Perceptron?
- ✓ Learning Rate
- ✓ Epoch
- ✓ Batch Size

Module-2: Tensorflow 2.0 with Tensor Board

- ✓ Introduction to Tensorflow 2.x
- ✓ Installing Tensorflow 2.x
- ✓ Introduction to TensorBoard
- ✓ Defining Sequence model layers
- ✓ Activation Function
- ✓ Layer Types
- ✓ Model Compilation
- ✓ Model Optimizer
- ✓ Model Loss Function
- ✓ Model Training
- ✓ Digit Classification using Simple Neural Network in Tensorflow 2.x
- ✓ **Mini project**

Module-3: Text Processing Methods

- ✓ Bag of Words
- ✓ Countvectorizer
- ✓ Term Frequency (TF)
- ✓ Inverse Document Frequency (IDF)
- ✓ Converting text to features and labels
- ✓ Multinomial Naive Bayes Classifier
- ✓ Leveraging Confusion Matrix Assignment
- ✓ Word Embeddings
- ✓ Word2Vec

Generative AI Specialization & ChatGPT, Prompt Engineering, Explainable AI

Module-1: Generative AI Specialization

- ✓ Introduction to Generative AI Models
- ✓ The Future of Generative AI
- ✓ Types of Generative AI Models
- ✓ Ethical Considerations in Generative AI Models & ChatGPT
- ✓ Popular Generative AI Models
- ✓ Benchmarking & Evaluating Models

Module-2: ChatGPT, Prompt Engineering, Explainable AI

- ✓ ChatGPT
- ✓ Fine-tuning ChatGPT
- ✓ Deploying and Scaling ChatGPT
- ✓ Maintaining ChatGPT
- ✓ Security and Privacy Considerations
- ✓ Monitoring and Debugging ChatGPT
- ✓ Mini project

Explainable AI

Prompt Engineering

Advanced Prompt Engineering Techniques

Edge AI

Module-1: Edge AI

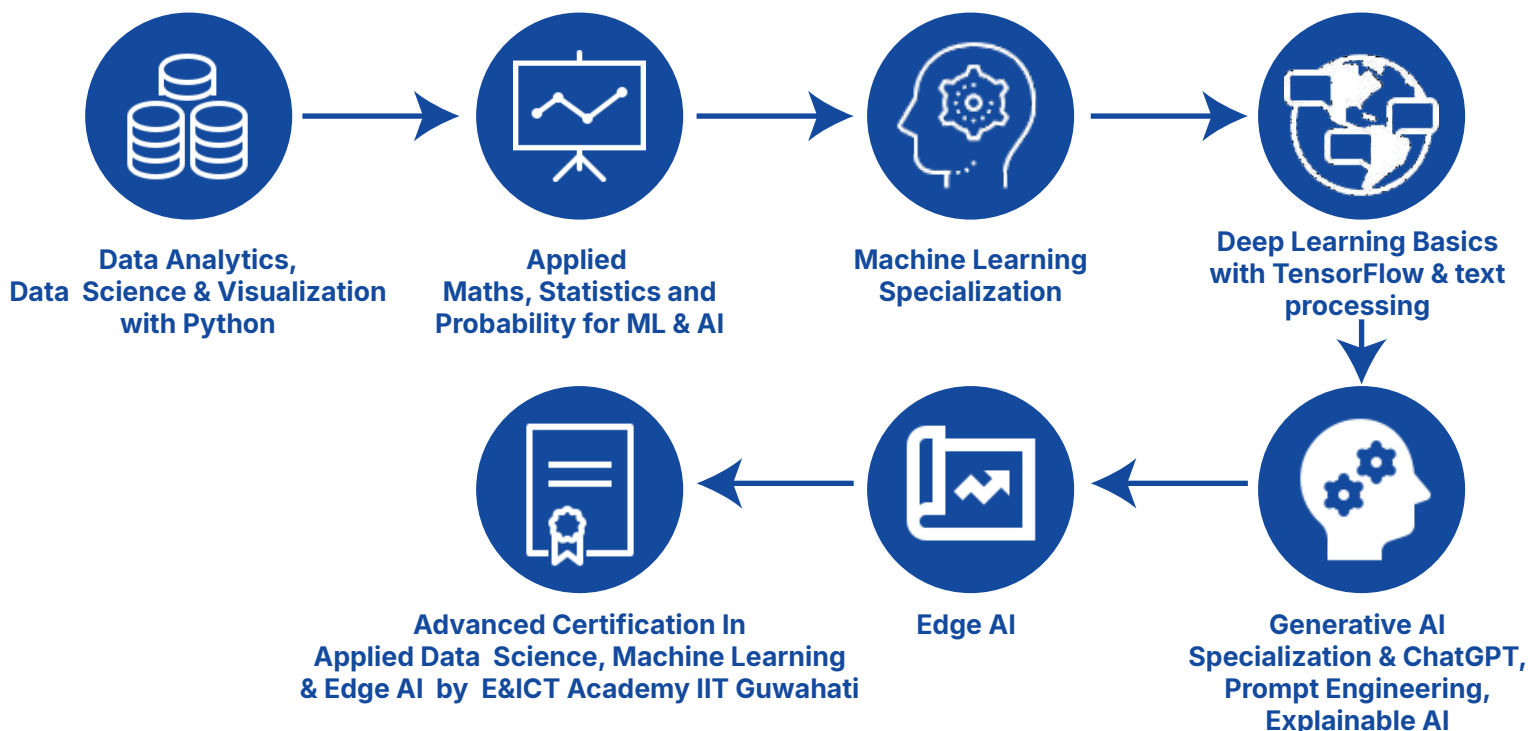
- ✓ Edge vs Cloud Computing
- ✓ Motivation behind Edge AI
- ✓ Edge AI applications
- ✓ Implementation process of Algorithms
- ✓ Inferences deployment on Edge

Major Capstone Project

TOOLS, LANGUAGES & FRAMEWORKS

				IP[y]:

LEARNING PATH



CONTACT



Program Information:

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By E&ICT Academy, IIT Guwahati



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